SOFTWARE QUALITY ASSURANCE / MASTER TEST SPECIFICATION

Piyush Portfolio Detailed QA Test Cases

117 detailed, executable test cases covering current behavior, identified risks, proposed recruiter-focused improvements, regression, accessibility, security, performance and release readiness.

117	10	14
Test cases	**Risk / defect entries**	**Traceability objectives**

DOCUMENT PURPOSE

This document turns the existing test inventory into one continuous execution register. Every case includes its objective, prerequisites, step-by-step procedure, acceptance criteria, priority, intended viewport/platform, evidence guidance, and execution fields. The added procedure guidance is intended to make execution repeatable and prevent code inspection from being mistaken for a functional pass.

SYSTEM UNDER TEST

Production URL	https://portfolio-135c5.firebaseapp.com/
Source repository	https://github.com/Piyush4521/piyush_portfolio
Implementation	Vite-powered frontend using HTML, JavaScript, CSS, JSON project data, Firebase Hosting and PWA/service-worker behavior.
Test approach	Read-only production smoke inspection plus controlled functional, visual, accessibility, security, performance and regression checks in local/staging where required.
Baseline date	12 October 2026. Desktop was inspected at approximately 1440 x 1000. Mobile viewports and several interactions remain execution gaps.

IMPORTANT STATUS RULE

All 117 cases are Not Run by default. The earlier desktop inspection is baseline context only. Do not mark a case Pass from the presence of an element, a source-code listener or an inferred outcome. A Pass requires the listed steps to be executed and evidence to be recorded.

READING THE REGISTER

Each case is headed by its unique ID and priority. The status starts as Not Run; fill actual result, defect ID and evidence only after execution. The appendices contain the baseline risk register and release gates; risks are not represented as confirmed defects unless reproduced.

Test Execution Policy & Priority Guide

Priority definitions

Priority	Meaning	Typical disposition
P0	Release blocker: core path failure, severe accuracy/security issue, critical accessibility barrier or build failure.	Must pass or be fixed before release; unresolved case needs explicit approval.
P1	Major functional, responsive, accessibility, credibility or resilience issue.	Fix before general release unless documented risk acceptance exists.
P2	Moderate content, usability or engineering quality issue with a narrower impact.	Track and resolve according to risk and release scope.
P3	Minor polish or low-impact issue.	Resolve within the planned quality backlog.

Execution status definitions

Status	Use when	Required record
Not Run	No tester has executed the steps yet.	No pass/fail claim; preserve as default.
Pass	All steps were executed and all acceptance criteria were observed.	Environment, actual result and evidence reference.
Fail	At least one expected result was not met.	Actual vs expected, reproduction steps, severity, defect ID and proof.
Blocked	The test could not be run due to environment, access, test data or dependency.	Precise blocker, attempted steps and unblock action.
N/A	The case is not applicable to the release/test target for a documented reason.	Reason and approver; do not use to hide untested scope.

Minimum execution evidence

Environment: Record browser/version, OS/device, exact viewport, deployed URL/build or commit, cache state, and network/CPU profile for relevant tests.

Actual behavior: Record the exact result observed, including URL/hash, focus, scroll, console/network behavior and all acceptance criteria that passed or failed.

Visual proof: Screenshots must show the full relevant state and be named with the test ID and viewport; reject loading, blank or wrong-state screenshots.

Defect lifecycle: Create an ID for each actionable failure; link the case; after the change, retest the failed case and linked regression cases.

Security discipline: Use harmless fixtures against local/staging only for injection/security tests. Do not run test payloads against production.

Release decision rule

Do not release while a P0 case is failed or untested, a P0/P1 defect remains open without documented acceptance, or required mobile/accessibility/security evidence is missing.

Execution Guide & Coverage Map

How to execute each test

1. Pin the build Record environment, deployment/build commit, browser version, operating system, viewport size, date, and cache/network conditions relevant to the case.
2. Use the exact steps Perform the steps in order. Add no assumptions to expected behavior. If preconditions are not met, mark Blocked or Not Run and explain why.
3. Compare with acceptance criteria Record actual UI/runtime outcome against every acceptance point; one missed acceptance point means Fail unless the test is formally split.
4. Capture reproducible evidence Save a screenshot, browser console/network log, accessibility output, PDF metadata, Lighthouse report, or approved source-of-truth reference as appropriate.
5. Log and retest defects For a failure, assign a defect ID and severity, include minimal reproduction steps, expected vs actual result, environment and evidence. After a fix, rerun the original case plus related regression cases.
6. Preserve release safety Run injection/security fixtures only in local or staging. Do not trigger real email/phone actions or deploy production changes as part of test execution.

Coverage groups

Area	Cases	Scope
Accessibility	9	Landmarks, contrast, keyboard, dialogs, reduced motion, targets and screen readers.
Command Palette	9	Open/search/select/no-result, keyboard, project and document destinations.
Contact	4	Email/phone/social targets and avatar behavior.
Contact / Legal	3	Privacy and terms dialogs plus policy accuracy.
Experience	5	Internship facts, PDF validity, case study and resume consistency.
Hero / Profile	7	Portrait, brand, role content, audio, install and social destinations.
Lab / Repositories	1	Repository card targets and accuracy.
Metrics / Content	9	Fallback display, animation, academic claims, counts and project truthfulness.
Navigation	8	Section routing, sticky/active state, keyboard path and external links.
Performance	4	Lighthouse, Web Vitals, asset weight, network/CPU throttling.
Projects / Dialog	8	Case-study details, close/focus, responsive dialog and injection resistance.
Projects / Reel	6	Card inventory, carousel, pause/drag/touch and boundaries.
PWA / Offline	7	Manifest, service worker updates, online/offline, install and cache.
Regression / Build	6	Build, deployment paths, release safety and post-change regression.
Regression / Content	1	Targeted behavior and acceptance checks listed within each case.
Responsive Layout	16	Nine viewport baselines, overflow, reflow, touch, orientation and zoom.
Security	4	Secrets, URL safety, dynamic HTML and privacy correctness.
SEO / Metadata	2	Canonical/OG metadata, robots and sitemap.
Smoke / Load	5	Page availability, assets, console/runtime and section rendering.
Tech Stack	3	Category filters, flip cards and keyboard semantics.

Total test cases: 117. All cases are presented in one continuous register below, grouped by ID and including evidence fields.

SMK-001	Open live portfolio root	P0

Area	Smoke / Load	**Type**	Smoke
Viewport / Platform	Desktop Chrome	**Status**	Not Run

PRECONDITIONS / SETUP	Network available; use production URL.
DETAILED TEST PROCEDURE	1. Navigate to https://portfolio-135c5.firebaseapp.com/. 2. Wait for page and images to settle. 3. Record final URL and load state. 4. Record current deployed URL/build identifier and test timestamp. 5. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	HTTP page renders the full portfolio without Firebase error page, blank screen, loading overlay stuck, or navigation loop.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SMK-002	Check first viewport assets	P1

Area	Smoke / Load	**Type**	Visual
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	SMK-001 passed.
DETAILED TEST PROCEDURE	1. Inspect hero portrait layers, logo/brand, hero text, social icons, and header controls. 2. Inspect failed image requests in browser devtools. 3. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 4. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 5. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Required assets load; no broken-image glyphs; image alt text is meaningful; layout remains usable if remote fonts fail.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SMK-003	Check browser console and runtime errors	P1

Area	Smoke / Load	**Type**	Runtime
Viewport / Platform	Desktop Chrome	**Status**	Not Run

PRECONDITIONS / SETUP	Live site loaded; DevTools console available.
DETAILED TEST PROCEDURE	1. Reload with console open. 2. Scroll through all sections. 3. Exercise representative controls. 4. Capture uncaught exceptions and failed critical requests. 5. Record current deployed URL/build identifier and test timestamp. 6. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No uncaught JavaScript errors, repeated exceptions, failed first-party assets, or unhandled promise rejections during the pass.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SMK-004	Verify canonical and hosted URL behavior	P1

Area	Smoke / Load	**Type**	SEO / Routing
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Public site accessible.
DETAILED TEST PROCEDURE	1. Open Firebase Hosting URL. 2. Inspect canonical metadata. 3. Open intended custom domain if configured. 4. Compare redirect/canonical strategy. 5. Record current deployed URL/build identifier and test timestamp. 6. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Production canonical URL is intentional, reachable, HTTPS, and consistent with the chosen primary domain; no contradictory domain metadata.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SMK-005	Verify all top-level sections render	P1

Area	Smoke / Load	**Type**	Smoke
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	SMK-001 passed.
DETAILED TEST PROCEDURE	1. Scroll from hero to footer. 2. Confirm About, Metrics, Work, Experience, Stack, Lab, Contact and Footer exist. 3. Record missing or duplicated sections. 4. Record current deployed URL/build identifier and test timestamp. 5. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Every advertised section is present once, has its expected content, and is not hidden behind an unintended overlay.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-001	Render at Small mobile (320 × 740)	P0

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	320 × 740	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-002	Render at Common Android mobile (360 × 800)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	360 × 800	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-003	Render at Modern mobile (390 × 844)	P0

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	390 × 844	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-004	Render at Large mobile (430 × 932)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	430 × 932	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-005	Render at Tablet portrait (768 × 1024)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	768 × 1024	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-006	Render at Tablet / small laptop (1024 × 768)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	1024 × 768	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-007	Render at Laptop (1366 × 768)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	1366 × 768	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-008	Render at Desktop baseline (1440 × 1000)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	1440 × 1000	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-009	Render at Large desktop (1920 × 1080)	P1

Area	Responsive Layout	**Type**	Responsive / Visual
Viewport / Platform	1920 × 1080	**Status**	Not Run

PRECONDITIONS / SETUP	Use responsive device emulation or a physical device; clear cache for first load.
DETAILED TEST PROCEDURE	1. Set the target viewport. 2. Reload the page. 3. Inspect hero, navigation, headings, metrics, project reel, experience, stack, contact and footer. 4. Capture screenshot. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No unintended horizontal page overflow, clipped content, collapsed essential controls, overlapping text, unreadably small text, or unreachable interactions at this viewport.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-010	Detect document-level horizontal overflow	P0

Area	Responsive Layout	**Type**	Responsive / Layout
Viewport / Platform	All viewports	**Status**	Not Run

PRECONDITIONS / SETUP	Open each viewport in RWD-001 through RWD-009.
DETAILED TEST PROCEDURE	1. Run `document.documentElement.scrollWidth` versus `clientWidth`. 2. Inspect `body`, `.stage`, `.browser-frame`, `.chapter`, reel, dialogs and tech grid. 3. Scroll to the far right if overflow exists. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Document-level width does not exceed viewport except for intentionally horizontally scrollable components; no accidental page-wide horizontal scrollbar.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-011	Reflow long headings and project names	P1

Area	Responsive Layout	**Type**	Visual
Viewport / Platform	Mobile + Tablet + Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Use 320px, 390px, 768px, and 1440px widths.
DETAILED TEST PROCEDURE	1. Inspect all display headings and long project titles. 2. Check the names KH147-TechOrbit, AI Stroke Detection System and OneMeal. 3. Confirm words are not cut off. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Text wraps naturally; meaningful words are not clipped; no mid-word breakage unless designed; hierarchy remains legible.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-012	Hero header controls fit on narrow screens	P1

Area	Responsive Layout	**Type**	Responsive / Functional
Viewport / Platform	Small mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Open at 320×740 and 390×844.
DETAILED TEST PROCEDURE	1. Inspect brand, Core Engineering, audio toggle and Install App. 2. Test each visible control without zooming. 3. Check controls do not cover the portrait or each other. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No overlap or clipping; essential actions remain visible/tappable; optional controls may be hidden only if equivalent functionality is available and this is intentional.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-013	Mobile navigation discoverability	P0

Area	Responsive Layout	**Type**	Responsive / Navigation
Viewport / Platform	Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Open at 320px and 390px widths.
DETAILED TEST PROCEDURE	1. Inspect bottom/mobile navigation. 2. Verify Home, About, Work, Experience and Contact are available. 3. Scroll near the page bottom and ensure nav does not cover links or dialog actions. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Mobile navigation is visible, labelled, usable with touch and keyboard, and does not obscure important page content or focus targets.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-014	Project reel interaction on touch	P1

Area	Responsive Layout	**Type**	Responsive / Interaction
Viewport / Platform	Mobile touch	**Status**	Not Run

PRECONDITIONS / SETUP	Use a real touch device or touch emulation.
DETAILED TEST PROCEDURE	1. Swipe horizontally across the project reel. 2. Scroll vertically over the reel. 3. Tap a project action after swiping. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	The reel can be browsed horizontally; vertical page scrolling remains possible; taps are not misinterpreted as drags; focus/action targets stay reachable.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-015	Responsive behavior on orientation change	P1

Area	Responsive Layout	**Type**	Responsive
Viewport / Platform	Mobile portrait + landscape	**Status**	Not Run

PRECONDITIONS / SETUP	Open at 390×844 then rotate to landscape or emulate 844×390.
DETAILED TEST PROCEDURE	1. Observe hero height and header. 2. Scroll through sections. 3. Open and close a dialog in each orientation. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Content reflows after resize/orientation change without stale fixed dimensions, clipped modal content, or a trapped scroll state.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

RWD-016	Browser zoom at 200% and 400%	P1

Area	Responsive Layout	**Type**	Accessibility / Responsive
Viewport / Platform	Desktop zoom	**Status**	Not Run

PRECONDITIONS / SETUP	Desktop browser with zoom controls.
DETAILED TEST PROCEDURE	1. Set zoom to 200% then 400%. 2. navigate main sections and open the command palette/dialogs. 3. Check controls and text. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No essential information or control is lost; content reflows where required; page remains operable without two-dimensional page scrolling except inherently spatial content.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-001	Desktop section navigation targets	P0

Area	Navigation	**Type**	Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Desktop viewport; page at top.
DETAILED TEST PROCEDURE	1. Click About, Metrics, Work, Experience, Stack, Lab and Contact individually. 2. Confirm URL hash and final scroll position. 3. Check the primary happy path and at least one relevant boundary or recovery path. 4. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each navigation item lands on its matching section; no target is missing; sticky nav does not hide section headings.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-002	Sticky navigation threshold and release	P1

Area	Navigation	**Type**	Functional / Visual
Viewport / Platform	Desktop + Tablet	**Status**	Not Run

PRECONDITIONS / SETUP	Page loaded on desktop.
DETAILED TEST PROCEDURE	1. Scroll down past the hero threshold. 2. Confirm nav becomes sticky. 3. Scroll back to the top. 4. Resize viewport while sticky. 5. Check the primary happy path and at least one relevant boundary or recovery path. 6. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Sticky state activates and clears predictably, does not flicker, and retains usable links without overlapping content.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-003	Active-section highlight matches scroll position	P2

Area	Navigation	**Type**	Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Page loaded; IntersectionObserver supported.
DETAILED TEST PROCEDURE	1. Scroll slowly through every section. 2. Observe active nav link. 3. Reverse direction and repeat. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Exactly the link for the visible section is marked active, with no stale or multiple active items except an explicitly designed state.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-004	Mobile navigation targets	P0

Area	Navigation	**Type**	Functional / Responsive
Viewport / Platform	Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Mobile viewport; mobile nav visible.
DETAILED TEST PROCEDURE	1. Tap Home, About, Work, Experience and Contact. 2. Check each destination and visible active/focus state. 3. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 4. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 5. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each mobile nav link reaches the expected target and remains usable after scrolling; no target is obstructed by the mobile nav.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-005	Skip link works using keyboard	P1

Area	Navigation	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Fresh page load; keyboard available.
DETAILED TEST PROCEDURE	1. Press Tab before interacting. 2. Verify skip link becomes visible. 3. Press Enter. 4. Inspect focus/scroll destination. 5. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 6. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Skip link is visibly revealed on focus and moves focus/navigation to main content rather than a non-content or missing anchor.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-006	Keyboard-only section access	P0

Area	Navigation	**Type**	Accessibility / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Fresh page load.
DETAILED TEST PROCEDURE	1. Use Tab / Shift+Tab to traverse links and controls. 2. Activate links with Enter and buttons with Enter/Space. 3. Confirm no keyboard trap. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	All interactive controls are keyboard reachable, have visible focus, and activate with expected keys; focus order follows reading order.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-007	URL fragments and browser back/forward	P2

Area	Navigation	**Type**	Navigation / Regression
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Use section links to navigate across at least three sections.
DETAILED TEST PROCEDURE	1. Navigate to Work, Experience, then Contact. 2. Use browser Back and Forward. 3. Reload at a section URL. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Fragment navigation and history behavior are consistent; no dead hash, stale scroll position, or blank view.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

NAV-008	External link safety attributes	P1

Area	Navigation	**Type**	Security / Links
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Page source / DOM accessible.
DETAILED TEST PROCEDURE	1. Inspect each external link. 2. For links opening a new tab, inspect target and rel. 3. Check URL scheme. 4. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 5. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	External destinations use HTTPS where expected; `target=_blank` links include `rel=noopener` (and noreferrer where appropriate); no unsafe/inert placeholder destination remains.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-001	Portrait color-reveal interaction	P2

Area	Hero / Profile	**Type**	Interaction
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Page at hero; portrait loads.
DETAILED TEST PROCEDURE	1. Activate portrait with click/tap. 2. Activate again. 3. Repeat by keyboard if button is focusable. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Color-reveal toggles consistently, remains labelled for assistive technology, and the portrait is not accidentally dragged/selected instead of activated.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-002	Portrait and profile image alt text	P1

Area	Hero / Profile	**Type**	Accessibility
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	DOM inspector available.
DETAILED TEST PROCEDURE	1. Inspect portrait and avatar img alt values. 2. Verify decorative layers are hidden from AT appropriately. 3. Run accessibility checker. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Informative images have useful alt text; duplicate/decorative layers are not announced redundantly; broken images have a meaningful fallback.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-003	Profile text matches approved role and experience	P0

Area	Hero / Profile	**Type**	Content / Regression
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Content owner has confirmed latest role dates and job title.
DETAILED TEST PROCEDURE	1. Compare headline, bio, internship title, location and availability with approved facts. 2. Verify status/date language is current. 3. Check on mobile and desktop. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Role claims, experience dates, location and availability are factual, current and consistently labelled across hero, About, badge and contact area.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-004	Core Engineering destination	P1

Area	Hero / Profile	**Type**	Functional / Links
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Core Engineering link visible.
DETAILED TEST PROCEDURE	1. Click/tap Core Engineering. 2. Confirm target URL/section loads. 3. Return using browser Back. 4. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 5. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Link leads to the intended embedded/IoT profile or clearly labelled destination, not a misleading unrelated anchor or a missing route.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-005	Audio toggle is optional and accurately labelled	P1

Area	Hero / Profile	**Type**	Accessibility / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Audio toggle visible; browser supports Web Audio or fallback.
DETAILED TEST PROCEDURE	1. Inspect initial toggle label/state. 2. Activate toggle twice. 3. Reload and inspect persisted state. 4. Check site remains fully usable with audio off. 5. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 6. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Audio is not required to use controls; label and actual mute state agree; preferences persist only as intended; unsupported audio APIs do not crash the page.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-006 Install App button only offers supported install — P1

Area	Hero / Profile	**Type**	PWA / Functional
Viewport / Platform	Desktop Chrome + Mobile Chrome	**Status**	Not Run

PRECONDITIONS / SETUP	PWA-compatible browser, plus browser with no install prompt.
DETAILED TEST PROCEDURE	1. Load in browser that supports beforeinstallprompt. 2. Load in browser that does not expose install prompt. 3. Inspect visibility and activation of Install App. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Install action appears only when it can be used or gives a clear fallback; clicking without a deferred prompt is not a dead/ambiguous primary action.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

HERO-007 Hero action/social destinations are meaningful — P1

Area	Hero / Profile	**Type**	Links / Content
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Page at hero.
DETAILED TEST PROCEDURE	1. Inspect GitHub, LinkedIn and any other social profile URL. 2. Click safe profile links and confirm correct account. 3. Do not activate mailto/tel links during automated run. 4. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 5. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each visible social link goes to the user's actual profile; no generic homepage placeholder such as `https://x.com` is presented as a personal account.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-001	Initial counter values are never misleading	P0

Area	Metrics / Content	**Type**	Content / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Load page with slow CPU/network or JavaScript disabled for comparison.
DETAILED TEST PROCEDURE	1. Inspect metric values before animation. 2. Reload and capture first 1 second. 3. Compare with final displayed values. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Metrics use meaningful approved fallback values or a neutral placeholder; no transient zeros/0.00 are presented as real achievement values.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-002	Counter animation reaches approved target	P1

Area	Metrics / Content	**Type**	Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Counters visible; motion allowed.
DETAILED TEST PROCEDURE	1. Scroll metrics into view. 2. Record each starting value and completed value. 3. Wait for animation to finish. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each counter finishes at exactly its approved target, precision and suffix; no NaN, duplicate suffix, rounding error or early termination.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-003	Academic metric label and value match official record	P0

Area	Metrics / Content	**Type**	Data Accuracy
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Approved transcript/grade sheet available to tester.
DETAILED TEST PROCEDURE	1. Compare displayed value (currently source target 8.43; label says Cumulative SGPA) with official record. 2. Verify whether it is cumulative, semester-specific or another aggregate. 3. Verify semester coverage. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Number, label, number format and coverage period all match the official grade record; do not publish a semester SGPA under a cumulative label.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-004	Hackathon wins/finalist count has supporting evidence	P1

Area	Metrics / Content	**Type**	Content / Evidence
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Certificates, result posts or event pages available.
DETAILED TEST PROCEDURE	1. Verify 1× Grand Winner and 2× Top 4 claim. 2. Confirm events, categories and result wording. 3. Open evidence links if provided. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each result is supported and accurately described; counts do not double-count the same achievement; link or proof is available where practical.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-005	Project count matches actual projects represented	P1

Area	Metrics / Content	**Type**	Data Accuracy
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Work section and portfolio JSON available.
DETAILED TEST PROCEDURE	1. Count unique project IDs/cards. 2. Compare with displayed 9+ or 9 number. 3. Check duplicates and spelling variants. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Displayed count and label accurately describe listed projects; prototypes are not misrepresented as production systems; duplicate projects are excluded.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-006	Absolute reliability and performance claims are substantiated	P0

Area	Metrics / Content	**Type**	Content / Credibility
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Project evidence and benchmark records available.
DETAILED TEST PROCEDURE	1. Search for 100%, zero latency, sub-100ms, 99.9%, verified and production claims. 2. Map each claim to test artifacts or remove/rephrase unsupported claims. 3. Verify project case study and home metric copy agree. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Absolute quantitative claims have reproducible evidence, scope and methodology; otherwise wording is rewritten as a design goal, prototype capability or qualitative description.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-007	Project descriptions match deployed implementation	P0

Area	Metrics / Content	**Type**	Content / Technical Accuracy
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Repositories and demo URLs available.
DETAILED TEST PROCEDURE	1. Compare features claimed for KH147/CHORD, OneOps, OneMeal and other projects with their repository and live demo. 2. Distinguish prototype from planned/future hardware integration. 3. Check all listed technology names. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Claims accurately distinguish implemented, simulated, planned and future integrations; no feature is claimed solely because it appears in a roadmap.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-008	Spelling and naming consistency	P2

Area	Metrics / Content	**Type**	Content / Quality
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Content pages, JSON and project cards accessible.
DETAILED TEST PROCEDURE	1. Check OneGuard/OneGaurd, KH147-TechOrbit/CHORD, capitalization and acronym definitions. 2. Compare titles between card, modal, repo and live demo. 3. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 4. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Product/project names are consistent with authoritative repository names; misspellings are corrected or intentional alias explained.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

MET-009	No-script and reduced-motion content remains available	P1

Area	Metrics / Content	**Type**	Accessibility / Resilience
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	JavaScript may be disabled; reduced-motion preference can be emulated.
DETAILED TEST PROCEDURE	1. Disable JavaScript and reload. 2. Enable `prefers-reduced-motion: reduce` and reload. 3. Inspect metrics and main content. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Core profile, project summaries, contact links and metric context remain accessible; animation is not required to discover information; reduced motion avoids unnecessary transitions/audio.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-001	All project cards render with expected fields	P0

Area	Projects / Reel	**Type**	Functional / Content
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Work section in view.
DETAILED TEST PROCEDURE	1. Inspect all nine current project entries. 2. Record card number, title, category, summary and technologies. 3. Compare with source data. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Every intended unique project card renders; no empty title, undefined label, malformed chip, duplicate or omitted primary project.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-002	Next arrow advances reel by the intended amount	P1

Area	Projects / Reel	**Type**	Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Work section in view.
DETAILED TEST PROCEDURE	1. Activate Next arrow repeatedly. 2. Observe card alignment and end behavior. 3. Repeat using keyboard. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Next advances exactly as designed, does not skip unexpectedly or strand cards; final-card behavior is deterministic; no console error.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-003	Previous arrow reverses reel	P1

Area	Projects / Reel	**Type**	Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Reel has been advanced beyond first card.
DETAILED TEST PROCEDURE	1. Activate Previous. 2. Repeat to first card. 3. Verify first-card boundary. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Previous reverses the reel consistently; first-card boundary does not produce blank space or an invalid transform.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-004	Auto-glide pauses for intentional interaction	P1

Area	Projects / Reel	**Type**	Accessibility / Interaction
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Reel gliding active.
DETAILED TEST PROCEDURE	1. Hover a card and inspect status. 2. Focus a card control with keyboard. 3. Move pointer away / blur control. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Auto-glide pauses while a user interacts or focuses controls, resumes at a predictable point, and does not move the target away before activation.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-005	Mouse drag and click do not conflict	P1

Area	Projects / Reel	**Type**	Interaction
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Desktop with mouse.
DETAILED TEST PROCEDURE	1. Drag horizontally across reel. 2. Release over a card. 3. Click CASE STUDY and LIVE buttons normally. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Drag scrolls only when pointer is dragged beyond threshold; click on buttons still activates the button; text selection and accidental navigation are handled sensibly.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-006	Touch drag and vertical scroll coexist	P1

Area	Projects / Reel	**Type**	Responsive / Interaction
Viewport / Platform	Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Mobile touch emulation or physical mobile device.
DETAILED TEST PROCEDURE	1. Swipe reel horizontally. 2. Swipe vertically over reel. 3. Tap action at rest. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Horizontal reel is browseable on touch while vertical page scroll works; no gesture traps or accidental demo opens.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-007	Open case study for each project	P0

Area	Projects / Dialog	**Type**	Functional / Content
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Work section rendered.
DETAILED TEST PROCEDURE	1. Activate CASE STUDY for every project card one at a time. 2. Inspect title, summary, architecture, highlights, decisions, challenges, lessons, stack and actions. 3. Close between iterations. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	The correct project detail opens for each card; content belongs to selected project; no undefined values or prior project's stale details.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-008	Case study close button works	P0

Area	Projects / Dialog	**Type**	Functional / Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Any case study dialog is open.
DETAILED TEST PROCEDURE	1. Click/tap close button. 2. Confirm dialog disappears. 3. Repeat with keyboard focus on close control. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Dialog closes on activation and returns user to the previous page/reel state; focus returns to the element that opened it.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-009	Escape key closes dialog	P1

Area	Projects / Dialog	**Type**	Accessibility / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Any case study modal open.
DETAILED TEST PROCEDURE	1. Press Escape. 2. Verify modal closes and scrolling is restored. 3. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 4. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Escape closes the native dialog, focus is restored, body scroll returns to its previous state and no invisible backdrop remains.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-010	Backdrop click behavior is intentional	P2

Area	Projects / Dialog	**Type**	Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Case study modal open.
DETAILED TEST PROCEDURE	1. Click inside dialog content. 2. Click backdrop outside dialog. 3. Observe close behavior. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Inside clicks do not close the dialog; backdrop behavior matches intended design and is consistent with Privacy/Terms dialogs.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-011	Dialog focus management and screen-reader label	P0

Area	Projects / Dialog	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Any dialog open; keyboard or accessibility inspector available.
DETAILED TEST PROCEDURE	1. Inspect accessible name. 2. Tab through controls. 3. Check whether focus remains in dialog. 4. Close and inspect focus return. 5. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 6. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Dialog has a meaningful accessible label, keyboard focus is contained while open, controls are reachable, and focus returns to opener.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-012	Dynamic project content is safely rendered	P0

Area	Projects / Dialog	**Type**	Security
Viewport / Platform	Local test fixture	**Status**	Not Run

PRECONDITIONS / SETUP	Project JSON and modal rendering source available.
DETAILED TEST PROCEDURE	1. Review dynamic content inserted via `innerHTML`. 2. Test benign special characters in a local test fixture (quotes, ampersands, angle brackets). 3. Verify markup is displayed as text unless HTML is explicitly intended. 4. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 5. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	User/content-controlled strings cannot inject script/event handlers or malformed markup; values are escaped or inserted using safe DOM text APIs.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-013	Live demo and repository links are accurate	P0

Area	Projects / Dialog	**Type**	Links / Functional
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	All project records loaded.
DETAILED TEST PROCEDURE	1. Check each card’s LIVE and GITHUB destinations. 2. Verify status code/reachability without signing in. 3. Confirm repo/project pairing. 4. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 5. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each shown action goes to its own relevant destination; unavailable demos are not shown as live; destinations do not lead to placeholders or unrelated content.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PRJ-014	Project detail content works at mobile size	P0

Area	Projects / Dialog	**Type**	Responsive / Dialog
Viewport / Platform	Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Mobile viewport with case study available.
DETAILED TEST PROCEDURE	1. Open a long case study at 320px and 390px. 2. Scroll inside dialog. 3. reach bottom action links. 4. Close. 5. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 6. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 7. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Dialog fits viewport, content scrolls within expected container, no horizontal overflow or clipped close button/actions, background page does not scroll inadvertently.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

STK-001	All technology categories filter correctly	P1

Area	Tech Stack	**Type**	Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Stack section in view.
DETAILED TEST PROCEDURE	1. Record initial ALL count. 2. Select each category: Languages, Frontend, Backend, AI & ML, Cloud & DevOps, Databases. 3. Compare cards/counts. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Selected filter shows only matching cards; count reflects visible cards; ALL restores every card; empty categories are handled gracefully.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

STK-002	Technology cards reveal details accessibly	P1

Area	Tech Stack	**Type**	Interaction / Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Stack section in view.
DETAILED TEST PROCEDURE	1. Hover card on desktop. 2. Click/tap card on mobile. 3. Use keyboard focus/activation if possible. 4. Inspect front/back content. 5. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 6. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Flip/reveal interaction is predictable, does not obscure text, and content is reachable without hover; keyboard and touch alternatives work.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

STK-003	Technology filters are keyboard-operable	P1

Area	Tech Stack	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Stack section in view.
DETAILED TEST PROCEDURE	1. Tab through filter controls. 2. Activate each with Enter/Space. 3. inspect focus and selected-state semantics. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Filters are keyboard accessible and selected state is programmatically exposed (e.g. aria-pressed/appropriate control semantics).
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

LAB-001	Lab repository cards and links	P1

Area	Lab / Repositories	**Type**	Links / Content
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Lab section in view.
DETAILED TEST PROCEDURE	1. Inspect all repository entries. 2. Verify title/description. 3. Open representative GitHub links in a separate tab. 4. Check for duplicates or missing targets. 5. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 6. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each repo card accurately describes the target repository and opens the correct public URL; no stale, renamed or placeholder URLs.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

EXP-001	Internship title, dates and organization are accurate	P0

Area	Experience	**Type**	Content / Evidence
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Approved experience letter or contract available.
DETAILED TEST PROCEDURE	1. Compare title, organization, product/team name, start/end date and location to signed experience letter. 2. Check timeline, badge, case study and contact copy. 3. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 4. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Dates and designation are consistent with primary source; “verified” language is used only where documentation supports it.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

EXP-002	Resume PDF link opens valid document	P0

Area	Experience	**Type**	Links / Document
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Resume link present; no file download should be forced during automation.
DETAILED TEST PROCEDURE	1. Inspect `href`. 2. Send HEAD/metadata request or open in a new tab if safe. 3. Confirm success content type is PDF and filename is correct. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Target returns a valid PDF response without 404/403 or HTML error content; file opens/downloads according to browser behavior; no stale version.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

EXP-003	Internship letter link opens valid document	P0

Area	Experience	**Type**	Links / Document
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Experience letter link present.
DETAILED TEST PROCEDURE	1. Inspect target URL. 2. Verify status/content type as PDF metadata only. 3. Confirm document name and date correspond to current experience section. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	File target returns a valid PDF and does not accidentally serve a duplicate résumé or unrelated document.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

EXP-004	SDE case study dialog works and closes	P1

Area	Experience	**Type**	Functional / Dialog
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Experience section in view.
DETAILED TEST PROCEDURE	1. Activate “View SDE Case Study”. 2. Inspect details and action links. 3. close by button and Escape. 4. open again after scrolling. 5. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 6. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Correct internship case study appears and closes; focus and scroll are restored; no empty or stale content.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

EXP-005	Resume links are consistent across site	P1

Area	Experience	**Type**	Content / Links
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Resume appears in header/command palette/experience/contact as applicable.
DETAILED TEST PROCEDURE	1. Collect all resume hrefs. 2. Compare filenames and URLs. 3. Check which version is intended to be canonical. 4. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 5. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Every resume action serves the approved current résumé or clearly identifies a variant; there are no stale duplicate files with confusing names.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-001	Open palette from visible button	P0

Area	Command Palette	**Type**	Functional / Dialog
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Desktop header nav visible.
DETAILED TEST PROCEDURE	1. Activate command button. 2. Inspect dialog, focus and default list. 3. Exercise both the success path and relevant close/escape/no-results paths; inspect focus on open and after closing. 4. Repeat at a narrow viewport and with keyboard only when available.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Palette opens once, input receives focus, useful destinations are listed, underlying page is not interactable while modal is open.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-002	Open palette with Ctrl+K and Cmd+K	P1

Area	Command Palette	**Type**	Functional / Keyboard
Viewport / Platform	Desktop Windows + macOS	**Status**	Not Run

PRECONDITIONS / SETUP	Browser/OS supports the shortcut.
DETAILED TEST PROCEDURE	1. Press Ctrl+K. 2. Close. 3. Press Meta/Cmd+K where available. 4. Repeat while focused in input. 5. Check the primary happy path and at least one relevant boundary or recovery path. 6. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Shortcut opens and toggles the palette without navigating browser history or interfering with browser shortcuts outside intended scope.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-003	Search filters destinations correctly	P1

Area	Command Palette	**Type**	Functional
Viewport / Platform	Desktop + Mobile if palette available	**Status**	Not Run

PRECONDITIONS / SETUP	Palette open.
DETAILED TEST PROCEDURE	1. Type “OneMeal”. 2. type “resume”. 3. type a partial section name. 4. clear input. 5. Check the primary happy path and at least one relevant boundary or recovery path. 6. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Results update immediately, preserve correct targets, and search is case-insensitive where intended.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-004	No-match state is understandable	P2

Area	Command Palette	**Type**	UX / Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Palette open.
DETAILED TEST PROCEDURE	1. Type a unique nonsense query. 2. inspect empty result state. 3. clear query. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No-match state is clear, not a blank broken dialog; clearing restores options; no runtime error occurs.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-005	Selecting a section navigates and closes palette	P0

Area	Command Palette	**Type**	Functional
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Palette open.
DETAILED TEST PROCEDURE	1. Select Work. 2. verify palette closes and Work section is reached. 3. Repeat for Contact and Experience. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Selecting a section closes the palette, scrolls to the correct section and preserves correct focus/scroll behavior.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-006	Selecting a project opens its case study	P1

Area	Command Palette	**Type**	Functional / Regression
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Palette open.
DETAILED TEST PROCEDURE	1. Search/select OneOps. 2. confirm correct project detail dialog opens. 3. close dialog. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Project command resolves to the correct project opener; it does not fail silently or open the wrong project.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-007	Resume and experience document commands work	P1

Area	Command Palette	**Type**	Functional / Links
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Palette open.
DETAILED TEST PROCEDURE	1. Select the SDE resume command. 2. Repeat for experience letter. 3. verify target is valid and palette closes. 4. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 5. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Commands open the correct document URL, safely handle new-tab behavior, and do not use a stale PDF path.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-008	Escape closes and focus returns	P1

Area	Command Palette	**Type**	Accessibility / Dialog
Viewport / Platform	Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Palette open with input focused.
DETAILED TEST PROCEDURE	1. Press Escape. 2. inspect page focus and scroll. 3. Reopen using shortcut. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Dialog closes without trapped focus; focus returns to trigger or sensible location; re-opening works.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CMD-009	Palette works with reduced motion and small viewport	P1

Area	Command Palette	**Type**	Accessibility / Responsive
Viewport / Platform	Mobile + Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Reduced motion enabled; test at 320px and 390px if palette is available.
DETAILED TEST PROCEDURE	1. Open palette. 2. Search and select an item. 3. Inspect dialog dimensions and motion. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Palette remains functional without unnecessary spatial transition; search field/results fit viewport and remain scrollable.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-001	Email link has correct address	P1

Area	Contact	**Type**	Link Validation
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Contact section visible.
DETAILED TEST PROCEDURE	1. Inspect email link href without launching email client. 2. Compare with approved public contact address. 3. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 4. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	`mailto:` target contains the approved address, no whitespace/typo, and link text clearly identifies email.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-002	Phone link is well formed	P1

Area	Contact	**Type**	Link Validation
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Contact section visible.
DETAILED TEST PROCEDURE	1. Inspect tel href without launching dialer. 2. Confirm country code and number formatting. 3. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 4. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	`tel:` target contains correct digits and country code; visible label is accurate; no accidental wrong number.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-003	GitHub and LinkedIn links target correct profiles	P1

Area	Contact	**Type**	Link Validation
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Contact area and hero/social data available.
DETAILED TEST PROCEDURE	1. Inspect hrefs. 2. Open in separate tab only if safe. 3. Confirm profile identity and HTTPS. 4. Verify the exact target, response status, content type and destination identity without unnecessarily downloading or submitting anything. 5. For new-tab links, inspect rel attributes; do not activate mailto/tel links during automated runs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	All social actions lead to correct user accounts, open as intended and are not generic platform homepages.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-004	Contact avatar interaction remains optional	P2

Area	Contact	**Type**	Interaction / Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Contact section visible.
DETAILED TEST PROCEDURE	1. Hover, click/tap and keyboard-reach avatar card. 2. inspect whether card is incorrectly focusable if noninteractive. 3. test with reduced motion. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Interaction cannot block actual contact links; it does not trigger excessive sound unexpectedly; keyboard semantics match behavior.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-005	Privacy policy modal opens and closes	P1

Area	Contact / Legal	**Type**	Legal / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Footer visible.
DETAILED TEST PROCEDURE	1. Open Privacy Policy. 2. read content and links. 3. close with close button and Escape. 4. test backdrop behavior. 5. Exercise both the success path and relevant close/escape/no-results paths; inspect focus on open and after closing. 6. Repeat at a narrow viewport and with keyboard only when available.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Policy dialog shows nonempty accurate policy text; can be closed predictably; focus and background scroll are restored.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-006	Terms modal opens and closes	P2

Area	Contact / Legal	**Type**	Legal / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Footer visible.
DETAILED TEST PROCEDURE	1. Open Terms. 2. confirm title/body are nonempty. 3. close via button and Escape. 4. Exercise both the success path and relevant close/escape/no-results paths; inspect focus on open and after closing. 5. Repeat at a narrow viewport and with keyboard only when available.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Terms dialog content is present, relevant and readable; no close/control failure or focus trap.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

CNT-007	Privacy and Terms accurately reflect actual data collection	P1

Area	Contact / Legal	**Type**	Legal / Security
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Policy and runtime behavior available.
DETAILED TEST PROCEDURE	1. Identify analytics, cookies/local storage, service worker caching, audio preference and contact data handling. 2. Compare to privacy notice. 3. check consent behavior if applicable. 4. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 5. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Public privacy/terms copy accurately describes implemented processing and third-party services; no unsupported privacy promises or obsolete legal claims.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-001 Manifest is valid and icons load — P1

Area	PWA / Offline	**Type**	PWA / Technical
Viewport / Platform	Desktop Chrome + Android Chrome	**Status**	Not Run

PRECONDITIONS / SETUP	HTTPS live site; browser devtools available.
DETAILED TEST PROCEDURE	1. Fetch `/manifest.json`. 2. Validate required fields, name, start_url, display, theme/background colors and icons. 3. Verify icon URLs return images. 4. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 5. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Manifest parses and contains valid same-origin icon URLs, suitable sizes, and start URL; browser shows no installability warnings attributable to the manifest.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-002 Service worker registers without repeated page reload — P0

Area	PWA / Offline	**Type**	PWA / Regression
Viewport / Platform	Chrome desktop + Android Chrome	**Status**	Not Run

PRECONDITIONS / SETUP	Fresh browser context; service worker enabled.
DETAILED TEST PROCEDURE	1. Load and reload twice. 2. Inspect Application > Service Workers. 3. observe updatefound/controllerchange behavior. 4. record navigations/reloads. 5. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 6. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Worker registers once and updates cleanly; no infinite/repeated reload cycle; update prompt/refresh behavior is predictable and does not interrupt user input.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-003	Offline banner reflects connection state	P1

Area	PWA / Offline	**Type**	PWA / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Service worker installed and page loaded.
DETAILED TEST PROCEDURE	1. Simulate offline mode. 2. confirm banner appears. 3. restore online. 4. confirm banner disappears. 5. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 6. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Offline status is accurate and accessible; banner never remains stale after network restoration; it does not obscure controls permanently.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-004	Offline cached page remains readable	P1

Area	PWA / Offline	**Type**	PWA / Resilience
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	After an initial successful online visit.
DETAILED TEST PROCEDURE	1. Visit page online once. 2. enable offline mode. 3. reload. 4. test core text and links where expected. 5. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 6. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Offline fallback matches documented caching scope; no blank app or broken core layout; uncached external actions are presented as unavailable rather than falsely succeeding.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-005	New version activation and cache invalidation	P1

Area	PWA / Offline	**Type**	PWA / Release
Viewport / Platform	Staging	**Status**	Not Run

PRECONDITIONS / SETUP	Test environment or safe release with service-worker change.
DETAILED TEST PROCEDURE	1. Install existing worker. 2. deploy a new asset revision in staging. 3. revisit site. 4. inspect worker state and asset versions. 5. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 6. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	New version activates without double reload or stale JS/CSS mismatch; cache policy refreshes HTML/service worker and versions static assets safely.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-006	Install prompt lifecycle	P1

Area	PWA / Offline	**Type**	PWA / Functional
Viewport / Platform	Chrome + Android	**Status**	Not Run

PRECONDITIONS / SETUP	Compatible browser plus unsupported-browser comparison.
DETAILED TEST PROCEDURE	1. Start installable context. 2. wait for install prompt event. 3. dismiss and retry. 4. complete only in test profile if needed. 5. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 6. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Install button follows event lifecycle; dismissed prompt doesn't cause errors; installed app hides install button; unsupported browsers have clear graceful behavior.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PWA-007	Persistent audio preference storage is handled safely	P2

Area	PWA / Offline	**Type**	Resilience / Privacy
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Browser storage allowed; inspect storage.
DETAILED TEST PROCEDURE	1. Toggle mute setting. 2. reload page. 3. clear localStorage / run private context. 4. check startup. 5. Record navigator online/offline state, service worker lifecycle state and cache state before and after the action. 6. Repeat from a clean browser context and an already-installed/cached context where the test requires it.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Storage read/write failures or blocked storage do not break startup; saved preference and visible state agree; no sensitive data is stored unnecessarily.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-001	Semantic landmarks and heading hierarchy	P1

Area	Accessibility	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Run axe or Lighthouse accessibility scan; inspect DOM.
DETAILED TEST PROCEDURE	1. Check header/nav/main/footer landmarks. 2. Review heading levels from h1 to section headings. 3. scan duplicate IDs. 4. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 5. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Landmarks are meaningful; exactly one primary h1 is used appropriately; headings form a logical hierarchy; IDs are unique.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-002	Color contrast of text and controls	P1

Area	Accessibility	**Type**	Accessibility / Visual
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Use contrast analyzer on rendered pages.
DETAILED TEST PROCEDURE	1. Sample normal text, muted labels, tags, disabled controls, buttons, focus states. 2. compare contrast ratios against WCAG AA criteria. 3. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 4. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Normal text meets 4.5:1 contrast; large text and relevant UI boundaries meet applicable criteria; state is not conveyed by color alone.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-003	Visible focus on every interactive element	P0

Area	Accessibility	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Keyboard testing enabled.
DETAILED TEST PROCEDURE	1. Tab through every section, carousel arrows, tech filters, dialogs and footer. 2. capture focus state. 3. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 4. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Focus indicator is visible against all backgrounds and is not clipped by overflow containers or obscured by sticky/mobile navigation.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-004	Accessible names for icon-only buttons	P1

Area	Accessibility	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Accessibility tree available.
DETAILED TEST PROCEDURE	1. Inspect hero orbit, sound, portrait, carousel arrows, command palette and close buttons. 2. verify accessible name and state. 3. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 4. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Every icon-only control has concise accessible name; toggle controls expose state; hidden decorative icon glyphs are not redundantly announced.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-005	Dialog focus trap and focus return for every modal	P0

Area	Accessibility	**Type**	Accessibility / Functional
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Case study, command, privacy and terms dialogs can be opened.
DETAILED TEST PROCEDURE	1. Open each dialog. 2. tab beyond final control. 3. Shift+Tab from first control. 4. close and inspect focus. 5. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 6. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Each modal keeps keyboard focus within its surface while open, supports Escape where expected, and returns focus to its opener.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-006	Reduced motion is respected across the site	P0

Area	Accessibility	**Type**	Accessibility / Motion
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	OS/browser reduce-motion setting enabled.
DETAILED TEST PROCEDURE	1. Reload with reduced motion. 2. scroll sections. 3. use reel, counters, avatar, portrait parallax and spatial flight. 4. compare normal motion. 5. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 6. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Nonessential motion is removed or reduced; content remains discoverable; transitions do not cause dizziness or hide required information; autoplay/glide pauses appropriately.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-007	Touch target sizes and spacing	P1

Area	Accessibility	**Type**	Accessibility / Mobile
Viewport / Platform	Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Physical or emulated mobile viewport.
DETAILED TEST PROCEDURE	1. Measure controls in hero, mobile nav, tech filters, carousel arrows and modal close. 2. tap adjacent controls repeatedly. 3. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 4. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Interactive targets meet target-size guidance or have sufficient spacing; adjacent actions are not easy to activate accidentally.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-008	Screen-reader reading order and image announcement	P1

Area	Accessibility	**Type**	Accessibility
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	NVDA/VoiceOver or accessibility tree.
DETAILED TEST PROCEDURE	1. Navigate headings, landmarks, portrait, avatar, dialog and nav. 2. compare order with visible layout. 3. Perform the manual keyboard and accessibility-tree checks described in the test steps; do not rely only on the presence of ARIA attributes. 4. Record focus order, visible focus, accessible name/state, and whether the interaction can be completed without a pointer.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Reading order is meaningful; text is not skipped; decorative visuals are hidden; no duplicate announcement from layered images.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

A11Y-009	Text resizing and non-text zoom	P1

Area	Accessibility	**Type**	Accessibility / Responsive
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Set browser text-only zoom if supported and normal zoom 200%.
DETAILED TEST PROCEDURE	1. Increase text size. 2. navigate metrics, long cards, legal dialogs and mobile nav. 3. verify content remains present. 4. Check at the exact viewport listed in this case, then inspect one neighboring breakpoint above and below where applicable. 5. Confirm page-level scroll width is not greater than viewport width unless the only overflow is an explicitly intended carousel/grid. 6. Check that important controls are reachable without browser zoom and that long content can be read without clipping.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Content adapts without clipping, truncation or overlapping interactive controls; essential information does not depend on fixed pixel text sizing.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SEO-001	Title and meta description are accurate	P1

Area	SEO / Metadata	**Type**	SEO
Viewport / Platform	Production	**Status**	Not Run

PRECONDITIONS / SETUP	HTML head accessible.
DETAILED TEST PROCEDURE	1. Inspect title, description, author, Open Graph title/description/image, canonical and theme-color. 2. compare with current portfolio identity. 3. Check the primary happy path and at least one relevant boundary or recovery path. 4. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Metadata is unique, accurate, nonempty, not misleading, and canonical/social URLs match the intended public domain.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SEO-002	Robots and sitemap are valid	P2

Area	SEO / Metadata	**Type**	SEO / Routing
Viewport / Platform	Production	**Status**	Not Run

PRECONDITIONS / SETUP	Public robots.txt and sitemap.xml URLs accessible.
DETAILED TEST PROCEDURE	1. Fetch `/robots.txt` and `/sitemap.xml`. 2. validate syntax and absolute URLs. 3. check sitemap URLs respond successfully. 4. Check the primary happy path and at least one relevant boundary or recovery path. 5. Confirm there is no unintended side effect on the rest of the page and no new browser console error.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Search engine directives do not accidentally block intended portfolio pages; sitemap contains canonical accessible URLs only.
EVIDENCE TO CAPTURE	- Capture a screenshot or log showing the relevant state and record environment details. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SEC-001	No secrets or private credentials in client bundle/repository	P0

Area	Security	**Type**	Security
Viewport / Platform	Repository + Build	**Status**	Not Run

PRECONDITIONS / SETUP	Source code/repository access; scanning tools available.
DETAILED TEST PROCEDURE	1. Scan source and built assets for API keys, access tokens, passwords, service account files and private URLs. 2. classify intentionally public client keys separately. 3. review git history if needed. 4. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 5. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No secret credential or private key is shipped to client or repository; public Firebase/API config has proper domain/API restrictions and backend rules.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SEC-002	External target and dynamic URL validation	P1

Area	Security	**Type**	Security
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	DOM/source inspection available.
DETAILED TEST PROCEDURE	1. enumerate external anchors and dynamic links in project modal/command palette. 2. confirm protocol/domain allowlist for generated targets. 3. inspect rel attributes. 4. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 5. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Only intended http/https/mailto/tel schemes appear in expected controls; dynamic URLs cannot be injected from project data; new-tab links are isolated.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SEC-003	Content injection/XSS regression	P0

Area	Security	**Type**	Security / Regression
Viewport / Platform	Local staging	**Status**	Not Run

PRECONDITIONS / SETUP	Use local fixture/staging only; never inject test payload into production.
DETAILED TEST PROCEDURE	1. Add safe test fixture strings containing `<script>`, quotes, ampersands and event attributes. 2. render cards, palette results and modal content. 3. inspect execution and DOM. 4. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 5. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Fixture is rendered as text or sanitized; no script runs, no event handler attributes are created and no layout breaks.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

SEC-004	Privacy/legal content does not overpromise	P1

Area	Security	**Type**	Security / Legal
Viewport / Platform	Production	**Status**	Not Run

PRECONDITIONS / SETUP	Privacy and terms content loaded.
DETAILED TEST PROCEDURE	1. Compare privacy text with localStorage, audio, service worker and any external scripts. 2. confirm contact link behavior and analytics use. 3. Run destructive/security payload tests only against a local build or staging environment and use a benign test fixture. 4. Capture the rendered DOM and console result; confirm test strings remain inert and no unexpected attributes or script execution occurs.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Notices accurately represent functionality; no claim like “verified production codebases” is unsupported by actual evidence; terms do not contradict the application.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PERF-001	Lighthouse baseline on desktop and mobile	P1

Area	Performance	**Type**	Performance / Quality Gate
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Production-like build and Chrome; no extensions.
DETAILED TEST PROCEDURE	1. Run Lighthouse Performance, Accessibility, Best Practices and SEO for desktop and mobile. 2. save JSON/HTML reports. 3. compare against release threshold. 4. Record cold/warm cache, network throttling, CPU throttling, browser version and run date. 5. Save the full audit output; distinguish lab data from field data and repeat at least twice if a metric is unstable.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Record reproducible baseline; no critical audit failure. Target by default: Performance ≥90, Accessibility ≥95, Best Practices ≥95, SEO ≥95 unless documented constraints approve otherwise.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PERF-002	Core Web Vitals / layout shift	P1

Area	Performance	**Type**	Performance
Viewport / Platform	Mobile + Desktop	**Status**	Not Run

PRECONDITIONS / SETUP	Production telemetry or Lighthouse/Web Vitals tooling available.
DETAILED TEST PROCEDURE	1. capture LCP, CLS and INP in representative mobile session. 2. reload with cache cold and warm. 3. inspect shift during font, image and hero animations. 4. Record cold/warm cache, network throttling, CPU throttling, browser version and run date. 5. Save the full audit output; distinguish lab data from field data and repeat at least twice if a metric is unstable.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No disruptive layout shift; CLS target ≤0.1, LCP ≤2.5s and INP ≤200ms at 75th percentile when field data is available; lab vs field data is labelled accurately.
EVIDENCE TO CAPTURE	- Screenshot at the exact viewport; record viewport width/height and whether document-level overflow occurred. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PERF-003 Image loading and asset weight — P1

Area	Performance	**Type**	Performance
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Network panel; cold-cache profile.
DETAILED TEST PROCEDURE	1. inspect image formats/sizes. 2. confirm hero high-priority image loads early. 3. check below-fold images use lazy loading where appropriate. 4. measure transferred bytes. 5. Record cold/warm cache, network throttling, CPU throttling, browser version and run date. 6. Save the full audit output; distinguish lab data from field data and repeat at least twice if a metric is unstable.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Hero visual is not delayed by unnecessarily large assets; below-fold images are loaded sensibly; width/height reserve space to reduce CLS; responsive formats are used where useful.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

PERF-004 Slow network and CPU throttling — P1

Area	Performance	**Type**	Performance / Resilience
Viewport / Platform	Mobile profile	**Status**	Not Run

PRECONDITIONS / SETUP	DevTools network/CPU throttling available.
DETAILED TEST PROCEDURE	1. test Slow 4G and 4× CPU slowdown. 2. reload. 3. open the project reel/modal and command palette. 4. observe responsiveness. 5. Record cold/warm cache, network throttling, CPU throttling, browser version and run date. 6. Save the full audit output; distinguish lab data from field data and repeat at least twice if a metric is unstable.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Core content appears without waiting for long decorative effects; controls remain responsive; no long main-thread task prevents navigation or text access.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-001	Production build succeeds after changes	P0

Area	Regression / Build	**Type**	Build / Regression
Viewport / Platform	Local	**Status**	Not Run

PRECONDITIONS / SETUP	Local repository with supported Node/npm.
DETAILED TEST PROCEDURE	1. Run `npm ci`. 2. Run `npm run build`. 3. inspect output and warnings. 4. Record current deployed URL/build identifier and test timestamp. 5. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	Build exits with code 0; no unresolved imports, syntax errors, missing public assets or newly introduced build warnings; `dist/index.html` and assets exist.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-002	Built asset paths resolve from Firebase Hosting	P0

Area	Regression / Build	**Type**	Build / Hosting
Viewport / Platform	Local preview + Firebase staging	**Status**	Not Run

PRECONDITIONS / SETUP	Built `dist` output and Firebase config available.
DETAILED TEST PROCEDURE	1. preview the build using `npm run preview`. 2. check all local assets/PDFs/icons/manifest/service worker. 3. verify SPA rewrites do not mask missing static assets. 4. Record current deployed URL/build identifier and test timestamp. 5. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	All first-party links resolve in the built site; static files are served from correct case-sensitive paths; rewrites do not return index.html for a missing PDF/asset.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-003	No changes to live site before review gate	P0

Area	Regression / Build	**Type**	Release / Governance
Viewport / Platform	Repository + Production	**Status**	Not Run

PRECONDITIONS / SETUP	Before release.
DETAILED TEST PROCEDURE	1. Compare current production URL/build fingerprint with baseline. 2. confirm proposed patch is on a review branch. 3. verify release/deploy is approved. 4. Record current deployed URL/build identifier and test timestamp. 5. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No production deployment occurs before the QA suite passes or remaining risks are explicitly accepted; main branch remains protected by review.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-004	Cross-browser compatibility	P1

Area	Regression / Build	**Type**	Compatibility
Viewport / Platform	Chrome / Edge / Firefox / Safari / Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Access Chrome, Edge, Firefox and Safari/WebKit; Android Chrome and iOS Safari where possible.
DETAILED TEST PROCEDURE	1. Repeat smoke test, nav, dialogs, command palette, carousel and PDF links. 2. verify CSS features and storage/Audio API fallbacks. 3. Record current deployed URL/build identifier and test timestamp. 4. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No critical browser-specific failure; unsupported optional capabilities degrade gracefully and core project/resume/contact information remains available.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-005	Check case-sensitive routes and asset names	P0

Area	Regression / Build	**Type**	Routing / Build
Viewport / Platform	Firebase preview	**Status**	Not Run

PRECONDITIONS / SETUP	Linux/Firebase hosting preview.
DETAILED TEST PROCEDURE	1. Visit paths for resume, experience letter, images, favicon, manifest and service worker. 2. compare href case to filename case. 3. Record current deployed URL/build identifier and test timestamp. 4. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	All referenced paths match exact filename case and return expected MIME type; no resource works only on case-insensitive local filesystem.
EVIDENCE TO CAPTURE	- Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-006	Regression test original interactions after recruiter improvements	P0

Area	Regression / Build	**Type**	Regression
Viewport / Platform	Desktop + Mobile	**Status**	Not Run

PRECONDITIONS / SETUP	Updated code under test.
DETAILED TEST PROCEDURE	1. Run hero, nav, reel, case dialogs, tech filters, lab links, experience case study, contact, command palette, PWA and legal dialogs. 2. compare to baseline. 3. Record current deployed URL/build identifier and test timestamp. 4. Capture actual result for the path that was executed; code/listener presence alone is not sufficient evidence of a pass.
EXPECTED RESULT / ACCEPTANCE CRITERIA	New recruiter actions and style changes do not break pre-existing interactions, sections, PWA behavior, URLs or visual identity.
EVIDENCE TO CAPTURE	- Record pre-state, action performed, resulting state, URL/hash or scroll destination, and whether focus/scroll was restored. - Attach relevant console/network/CI/DevTools log or report; include exact browser, build commit and reproduction conditions. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

REG-007	Resume, metrics, project copy and CTAs remain consistent	P0

Area	Regression / Content	**Type**	Regression / Content
Viewport / Platform	All	**Status**	Not Run

PRECONDITIONS / SETUP	Updated code under test; content owner approval available.
DETAILED TEST PROCEDURE	1. Compare CTA destinations, resume variant, metric labels, project summaries and contact data across all locations. 2. search source and built output for stale copy. 3. Compare displayed text and numeric values to an owner-approved source of truth; record source/version and exact mismatch if any. 4. Do not infer truth from current portfolio copy alone. Claims that cannot be substantiated should remain open for owner review.
EXPECTED RESULT / ACCEPTANCE CRITERIA	No stale labels, duplicate/conflicting academic metrics, unsubstantiated claims, placeholder links or inconsistent résumé/document links remain after the patch.
EVIDENCE TO CAPTURE	- Cite the approved source of truth or file URL/version; note exact text/value checked without exposing private data. - If failed: create a defect ID, attach reproduction evidence, and do not mark Pass until the fix is retested.
ACTUAL RESULT / DEFECT	Execution field: not run. Record actual result, defect ID, exact environment and evidence reference after running this case.

Appendix A - Baseline Observations & Open Risks

These items are not all confirmed bugs. The register preserves their evidence classifications from the source QA workbook. Verify or resolve each item with evidence before release sign-off.

ID	Classification / Priority	Observation / Risk	Required action	Related tests
OBS-001	Confirmed observation / P2	Placeholder social destination. Hero/source shows an X icon that points to the generic https://x.com/ homepage rather than a user profile.	Replace with a verified user profile or remove the icon; confirm all social links map to the correct account.	NAV-008; HERO-007; CNT-003
RISK-002	Needs verification / P0	Academic metric label/value may be inconsistent. Source markup uses data-target=8.43 while label says “Cumulative SGPA” and describes coverage through Semester VI. Official grade sheet was not available in this browser run.	Check the approved transcript/grade sheet; define whether the value is semester-specific or cumulative; align label, value, precision and supporting text.	MET-003; MET-001; REG-007
RISK-003	Needs evidence review / P1	Absolute reliability/performance claims lack inline evidence. Copy includes absolute or quantitative language such as 100% reliability, zero-latency bottleneck, and sub-100ms performance claims. No supporting benchmark artifacts were examined in the live visual pass.	Attach reproducible test/benchmark evidence or rephrase claims as prototype capability, engineering approach or design goal.	MET-006; MET-007; SEC-004
GAP-004	Test coverage gap / P0	Mobile viewport coverage not executed. Initial browser QA could not resize to 390×844 or 320×740. No direct mobile visual pass was completed.	Run RWD-001 through RWD-016 on responsive emulation and preferably a physical Android/iOS device; capture screenshots.	RWD-001:RWD-016; A11Y-007; PERF-001
GAP-005	Test coverage gap / P0	Interactive behaviors not fully exercised. Initial pass primarily inspected visible elements/source listeners; command palette and dialogs were not all opened/clicked during that pass.	Execute NAV, PRJ, CMD, STK, EXP, CNT, PWA and A11Y test groups using real interactions.	PRJ-002:PRJ-014; CMD-001:CMD-009; PWA-001:PWA-007
RISK-006	Needs runtime verification / P1	Potential metric flash/zero fallback. Source markup originally contains `0` / `0.00` fallback text with animation targets; current runtime behavior was not captured in a video or controlled slow-load test.	Use MET-001 and MET-002 under cold cache, slow network, reduced motion and no-JS conditions; retain meaningful fallback content.	MET-001; MET-002; MET-009
RISK-007	Needs runtime verification / P1	PWA install/update behavior may be unclear. Install button, service worker registration, updatefound and controllerchange reload logic exist; supported/unsupported browser states were not exercised.	Run HERO-006 and PWA-001 through PWA-007, with special attention to dead install button and repeated reload during updates.	HERO-006; PWA-001:PWA-007
RISK-008	Needs security test / P0	Dynamic HTML rendering needs XSS regression coverage. Project data is inserted into modal and command-palette HTML. Existing text normalization alone does not prove HTML escaping/sanitization.	Review all dynamic HTML sinks and run SEC-003 on a local test fixture; use safe DOM text APIs or escaping where appropriate.	PRJ-012; CMD-003; SEC-002; SEC-003
RISK-009	Repository hygiene / P2	Repository hygiene: generated dependencies/build output tracked. Repository tree contains `node_modules` and `dist`. This can bloat repo and create source/build drift; check .gitignore and workflow before any cleanup.	Confirm intentional tracking policy; add appropriate ignore rules and remove generated dependency tree from version control only through a reviewed change.	REG-001; REG-002; REG-003
RISK-010	Needs owner review / P1	Canonical metadata vs Firebase URL requires decision. Page source uses canonical/Open Graph URL on a custom domain while the QA target is Firebase Hosting. This may be intentional if the custom domain is primary.	Confirm custom domain and Firebase URL redirect/canonical strategy; check HTTP status/redirects, sitemap and social preview URLs.	SMK-004; SEO-001; SEO-002

Appendix B - Release Exit Criteria

Gate	Required exit condition	Required evidence
Critical functional	All P0 cases pass; no open P0/P1 defect affecting release; blocked P0s have an approved disposition.	Execution sheet, reproduction evidence and retest record.
Responsive	Test at 320, 390, 768, 1024, 1440 and 1920 widths, plus mobile landscape; no unintended document overflow or inaccessible controls.	Screenshots labelled by viewport and device/browser.
Accessibility	Keyboard path, visible focus, dialog management, reduced motion and contrast checks completed; critical barriers fixed.	Manual notes plus automated accessibility report.
Content integrity	Academic values, internship title/dates, project capabilities and quantitative claims verified against owner-approved evidence.	Approved grade data, experience letter, benchmarks or explicit rewording decision.
Security	No shipped secrets; dynamic HTML sinks reviewed and XSS regression checked in local/staging; links validated.	Code review, scan output and benign fixture results.
Build / hosting	Clean install and production build pass; case-sensitive assets, PDF documents, manifest and service worker resolve correctly.	Build log, hosting preview and network checks.
Release safety	All planned changes reviewed; production deployment explicitly approved; smoke/regression passes after deploy.	PR/change approval, deployment ID and post-deploy smoke evidence.